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INTRO **About me:** I am a researcher working on the safety and alignment of artificial intelligence systems. My research interests lie in exploring pathways towards robust, stable and trustworthy AI. These explorations include principled theoretical and computational analysis of modern computer vision and large language models, their [structures](#), [properties](#) and methods of [optimisation](#). I have extensive experience in collaborations with applied mathematicians, archaeologists and medical clinicians, bridging frontier AI mathematical theory to practical AI applications. My current research focus is on safety and alignment through principled and deterministic intervention of the model with theoretical guarantees, e.g. providing methods to [stain](#), [lock](#), [edit](#) or [attack](#) AI models post-training with only minimal optimization required.

SELECT FIRST-AUTHOR PUBLICATIONS

- [Stealth edits for provably fixing or attacking large language models](#)

Oliver Sutton, Qinghua Zhou*, Wei Wang, Desmond Higham, Alexander Gorban, Alexander Bastounis, and Ivan Tyukin*

Advances in Neural Information Processing Systems (NeurIPS), Vancouver Canada, 2024

[Huggingface Demo](#) | [Code](#) | [SIAM News Front Page Article](#)

This work exposes the susceptibility of modern AI models to a new class of malicious attacks and reveals a new theoretical understanding of the causes behind this; when an attacker provides a specific prompt, the model will generate the attacker's desired outputs. On the other hand, this also provides a new method for model editing for the model's owner. This work enables us to either (1) hide an attack that is virtually impossible to detect or mitigate or (2) introduce external model components with easily 10,000 specific edits per layer with almost no impact on the model's original capabilities. The method works across all common architectural families of LLMs.

- [Staining and locking computer vision models without retraining](#)

Oliver Sutton, Qinghua Zhou*, George Leete, Alexander Gorban, and Ivan Tyukin*

International Conference on Computer Vision (ICCV), Hawai'i US, 2025

This work provides a method to (1) stain, i.e. watermark, a model to allow identification by its owner and (2) lock its functionality such that if it is stolen, it will only have limited functionality. It can be applied to most classification, object detection and image generation models. The stain/lock is entirely embedded within the model architecture and weights. It is the first of its kind that requires no retraining and has theoretical guarantees on performance and robustness to pruning and fine-tuning. This is a method that can be applied to CNNs, GANs, ViTs, diffusion models and more.

- [Taming robustness bias in large language models](#)

Under Review, 2025

This work attempts to stabilize LLM responses to non-semantic textual corruptions, e.g. typos; and CV models to nominal noise, e.g. weather, with little or no fine-tuning. This is done through the identification of a lower bound on robustness certification radius – a "robustness bias" describing systematic expected shift in outputs. Motivated by this analysis, practical techniques developed show that improvement to robustness (and sometimes even performance) can be achieved via simple targeted debiasing, e.g. by adding a small adapter or adjusting neuron biases.

- [Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data](#)

Qinghua Zhou, Shuihua Wang, Hengde Zhu, Xin Zhang, Yu-Dong Zhang

Neural Networks, (167) 380-399, 2023

A key problem of many practical applications is how to learn from small sample size datasets with high-dimensional representations. Here, we introduce a novel class of stacking methods that utilise attention pooling mechanisms for the construction of both ensembles and cascades. The method is empirically validated for its functionality and performance on a range of imaging datasets in the medical domain.

*Equal contributions, listed in alphabetical order.

OTHER WORKS

- [Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data](#)

Qinghua Zhou, Oliver Sutton, Yu-Dong Zhang, Alexander Gorban, Valeri Makarov, Ivan Tyukin

This work considers the scenario of mesoscopic sample-size, where we cannot reliably build models with high degrees of expressivity. This work presents a neuromorphic algorithm capable of improving existing feature spaces of pre-trained models via learning relevant associations in high-dimensional data. This is an extension of our prior work on the [theoretical foundations of few-shot learning](#).

- [Quasi-orthogonality and intrinsic dimensions as principled measures of learning and generalisation](#)

Qinghua Zhou, Alexandar Gorban, Jonathan Bac, Andrei Zinovyev, Ivan Tyukin

This work examines the correlations between the accuracies of trained networks and measures of randomly initialised networks. While prior works focused on volume-based principled measures, this work examines high-dimensional measures of intrinsic dimensionality and quasi-orthogonality. The observations validate [theoretical findings on separability](#), and provides a new perspective for zero-shot NAS.

- [How adversarial attacks can disrupt seemingly stable accurate classifiers](#)

Oliver Sutton, Qinghua Zhou, Ivan Tyukin, Alexandar Gorban, Alexander Bastounis, Desmond Higham

This work demonstrates a fundamental feature of classifiers on high-dimensional data: simultaneous susceptibility to adversarial attacks and robustness to random perturbations. This work provide theoretical & empirical verification that: using additive noise is inefficient for eradicating or detecting adversarial examples, and fundamentally not a good approach for certification of robustness.

- More publications can be found below, or on my [Google Scholar Page](#)

ACADEMICS

King's College London

2022–2025

Department of Mathematics, United Kingdom

- [Postdoctoral Research Associate in Methods and Algorithms of Artificial Intelligence](#).
- Topic: Robust, resilient, certifiable and trustworthy AI systems
- Supervisor: Prof. [Ivan Tyukin](#)

University of Leicester,

2019–2023

School of Computing and Mathematical Sciences, United Kingdom

- [Ph.D., Computer Science Research](#) with Graduate Teaching Assistant Studentship
- Topic: Disease detection in high-dimensional low sample size medical data
- Supervisors: Prof. Yudong Zhang and Prof. [Ivan Tyukin](#)

University of Sydney,

2016–2018

School of Mathematics and Statistics, and School of Physics, Australia

- [B.Sc., Science \(Advanced Mathematics\)](#)
- Majors in both Physics and Applied Mathematics
- Introduction to research through astero-seismology with Prof. [Tim Bedding](#), where I worked on calibration of Kepler light curves for MIA stars

OTHER EXPERIENCES

Industry Collaboration with TG-0

2024

Department of Mathematics, King's College London, and TG-0, London

- Development of reservoir computing algorithms for specific hardware applications
- Development of low-cost and efficient algorithmic alternatives for specialised hardware

Academic Teaching - Graduate Teaching Assistant

2019–2021

School of Computing and Mathematical Sciences, University of Leicester

- Teaching in a range of undergraduate to master’s modules.
- General student admin, tutorials, labs and examinations.

Research Assistant Coordinator

2019–2020

Collaboration with Department of Cardiovascular Sciences, University of Leicester.

- Study and processing of gene expression data for ML-based knowledge discovery
- Lead and coordinate a small group of research assistants in the same project.

GRANTS & ACCREDITA- TIONS

- **Lambda Research Grant** 2025: \$3000
- **Associate Fellow of the Higher Education Academy** (AFHEA)

SKILLS

- Proficient in Mathematica, MATLAB and Python
- Proficient in Tensorflow and Pytorch
- Proficient in machine learning, AI and data science frameworks for both CV and NLP
- Proficient in image processing frameworks both general (e.g., OpenCV, PIL) and specific (e.g. Nibabel, FSL).
- Beginner in formal mathematical languages for autoformalisation: Lean and ISabelle.

LIST OF PUBLICATIONS

- **Zhou, Q.**, Tyukin, I.Y. 2025. Taming robustness bias in large language models. Under Review.
- Sutton, O.J.*, **Zhou, Q.***, Leete, F., Gorban, A., Tyukin, I.Y. 2025. [Staining and locking computer vision models without retraining](#). In proceedings of *International Conference on Computer Vision (ICCV)*.
- Sutton, O.J.*, **Zhou, Q.***, Wang, W., Higham, D.J., Gorban, A.N., Bastounis, A., Tyukin, I.Y. 2024. [Stealth edits for provably fixing or attacking large language models](#). In proceedings of *Conference on Neural Information Processing Systems (NeurIPS)*. Online Demo: [huggingface-spaces/stealth-edits](#). + SIAM News [Headline Article](#).
- Sutton, O.J., **Zhou, Q.**, Tyukin, I.Y., Gorban, A.N., Bastounis, A. and Higham, D.J., 2024. [How adversarial attacks can disrupt seemingly stable accurate classifiers](#). *Neural Networks*, 180, 106711.
- Tyukin, I.Y., Tykina, T., van Helden, D.P., Zheng, Ze., Mirkes, E.M., Sutton, O., **Zhou, Q.**, Gorban, A.N., Allison, P., 2024. [Coping with AI errors with provable guarantees](#) *Information Sciences*, 678, 106711.
- Tyukin, I.Y., Tykina, T., van Helden, D.P., Zheng, Ze., Mirkes, E.M., Sutton, O., **Zhou, Q.**, Gorban, A.N., Allison, P. 2024. [Weakly Supervised Learners for Correction of AI Errors with Provable Performance Guarantees](#). In proceedings of *International Joint Conference on Neural Networks*.
- Zhu, H., Wang, W., Ulidowski, I., **Zhou, Q.**, Wang, S., Chen, H., Zhang, Y., 2023. [MEEDNets: Medical Image Classification via Ensemble Bio-inspired Evolutionary DenseNets](#). *Knowledge-Based Systems*, 280, 111035.
- **Zhou, Q.**, Wang, S., Zhu, H., Zhang, X., & Zhang, Y. (2023). [Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data](#). *Neural Networks*, 167, 380-399.
- Sutton, O., **Zhou, Q.**, Gorban, A.N., Tyukin, I.Y., 2023. [Relative intrinsic dimensionality is intrinsic to learning](#). In proceedings of *International Conference on Artificial Neural Networks (ICANN)*.

*Equal contributions, alphabetical order

- Bastounis, A., Gorban, A.N., Hansen, A.C., Higham, D.J., Prokhorov, D., Sutton, O., Tyukin, I.Y., **Zhou, Q.**, 2023. [The Boundaries of Verifiable Accuracy, Robustness, and Generalisation in Deep Learning](#). In proceedings of *International Conference on Artificial Neural Networks (ICANN)*.
- **Zhou, Q.**, Sutton, O., Zhang, Y.D., Makarov, V.A., Gorban, A.N., Tyukin, I.Y., 2023. [Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data](#). In proceedings with oral presentation in *International Joint Conference on Neural Networks*.
- **Zhou, Q.**, Gorban, A.N., Mirkes, E.M., Bac, J., Zinonyev, A., Tyukin, I.Y., 2022. [Quasi-orthogonality and intrinsic dimensions as measures of learning and generalisation](#). In proceedings with oral presentation in *International Joint Conference on Neural Networks*.
- **Zhou, Q.**, Wang, S., Zhang, X., Zhang Y.D., 2021. [WVALE: Weak Variational Autoencoder for Localisation and Enhancement of COVID-19 Lung Infections](#). *Computer Methods and Programs in Biomedicine*, 221, 106883. + Physicsworld [Front Page Article](#).
- Wang, S., Staphaty, S.C., **Zhou, Q.**, Zhang, X., Zhang, Y., 2022. [Secondary pulmonary tuberculosis identification via pseudo-Zernike moment and deep stacked sparse autoencoders](#). *Journal of Grid Computing*, 20, 1-16.
- Yu, X., **Zhou, Q.**, Wang, S. and Zhang, Y.D., 2021. [A systematic survey of deep learning in breast cancer](#). *International Journal of Intelligent Systems*, 37(1), 152-216.
- **Zhou, Q.**, Zhu, H., Zhang, X. and Zhang, Y., 2021, December. [Knowledge distillation for secondary pulmonary tuberculosis classification ensemble](#). In Proceedings of the *14th IEEE/ACM International Conference on Utility and Cloud Computing Companion*.
- Wang, S., Celebi, M.E., Zhang, Y.D., Yu, X., Lu, S., Yao, X., **Zhou, Q.**, Miguel, M.G., Tian, Y., Gorriz, J.M. and Tyukin, I., 2021. [Advances in Data Preprocessing for Biomedical Data Fusion: An Overview of the Methods, Challenges, and Prospects](#). *Information Fusion*, 76, 376-421.
- **Zhou, Q.**, Zhang, X. and Zhang, Y.D., 2021. [Ensemble learning with attention-based multiple instance pooling for classification of SPT](#). *IEEE Transactions on Circuits and Systems II*, 69(3), 1927-1931.
- Tyukin, I.T., Gorban, A.N., Alkhudaydi, M.H., and **Zhou, Q.**, 2021. [Demystification of Few-shot and One-shot Learning](#). In Proceedings of the *International Joint Conference on Neural Networks*.
- Wang, S.H., Wu, K., Chu, T., Fernandes, S.L., **Zhou, Q.**, Zhang, Y.D. and Sun, J., 2021. [SOSPCNN: Structurally Optimized Stochastic Pooling Convolutional Neural Network for Tetralogy of Fallot Recognition](#). *Wireless Communications and Mobile Computing*, 2021(1), 5792975.
- Chen, X., **Zhou, Q.**, Lan, R., Wang, S.H., Zhang, Y.D. and Luo, X., 2021. [Sensorineural hearing loss classification via deep-HLNet and few-shot learning](#). *Multimedia Tools and Applications*, 80(2), 2109-2122.
- Zhu, M., Zhao, D., Wang, Y., **Zhou, Q.**, Wang, S., Mo, X., Yang, M. and Sun, Y., 2020. [Multi-Slice Radiomic Analysis of Apparent Diffusion Coefficient Metrics Improves Evaluation of Brain Alterations in Neonates With Congenital Heart Diseases](#). *Frontiers in neurology*, 11, 586518.
- Zhang, Y.D., Dong, Z., Wang, S.H., Yu, X., Yao, X., **Zhou, Q.**, Hu, H., Li, M., Jiménez-Mesa, C., Ramirez, J. and Martinez, F.J., 2020. [Advances in multimodal data fusion in neuroimaging: Overview, challenges, and novel orientation](#). *Information Fusion*, 64, 149-187.
- Kang, C., Li, Y., Novak, D., Zhang, Y., **Zhou, Q.** and Hu, Y., 2020. [Brain Networks of Maintenance, Inhibition and Disinhibition During Working Memory](#). *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(7), 1518-1527.
- **Zhou, Q.**, Mirkes, EM., Zhang, Y., Tyukin, I., Gorban, AN., Yu, X., Wang S., Charlton M., Coats, T., Sims MR., Thompson, JP. 2020, Machine learning technique for severity classification in sepsis patients. *Intensive Care Medicine Experimental 2020*, 8(Suppl 2):73 ID 000473.
- **Zhou, Q.**, Zhu, H., Yao, X., Zhang, Y., Murphy, G., Adebayo, A., Wozniak, M., 2020. Obesity-driven differences in gene expression – a machine learning approach. Abstract at *SCTS 2020 conference*.